The Weinberg Angle from Hopf Bundle Geometry

A Hopf-Geometric Route to the Same Estimate — and the Same Gap

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Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it.

Abstract

This paper develops the Weinberg-angle estimate through the Hopf bundle structure that the rope two-strand state space carries, offering a second, more geometric route to $\sin^2\theta_W = 1/(3\sqrt{2})$. The Hopf framing makes the origin of the '3' and '√2' more transparent, but it reaches the same numerical value and therefore inherits the same ~1.94% gap from measurement. It is reported with the identical honest caveat: suggestive geometry, not a precision derivation. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants.

Scope of this paper

CLAIM: a Hopf-bundle route to $\sin^2\theta_W = 1/(3\sqrt{2})$, clarifying the geometric origin of the factors.

NOT CLAIMED: improved precision. Same ~2% gap as the winding route. Status: Modeled (Conjecture).

1. The Hopf Bundle of the Two-Strand State

The rope two-strand state space carries the Hopf bundle structure $S^3 \to S^2$. Reading the mixing angle from the base/fibre geometry of this bundle reproduces the winding-angle estimate, with the 3 arising from the spatial embedding and the √2 from the two-strand fibre.

2. Same Value, Same Gap

The Hopf route yields $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$, identical to the winding-geometry estimate and hence 1.94% from the measured value. The geometric clarity is the contribution; the numerical status is unchanged.

The honest status of this value. The predicted $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ differs from the measured 0.23122 by about 1.94%. It is close, and its form is suggestive, but it is NOT exact, and the test suite pins it explicitly as a 'soft external input' — the value that, when fed into the lepton-mass relation instead of the measured angle, breaks it. This is therefore a geometric conjecture of the right magnitude, not a precision derivation, and the ~2% gap is the known weak link of the electroweak sector.

Status

Hopf-geometric $\sin^2\theta_W = 1/(3\sqrt{2})$ — Modeled (Conjecture, same value as winding route).
~1.94% deviation — unchanged; the sector's weak link.

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